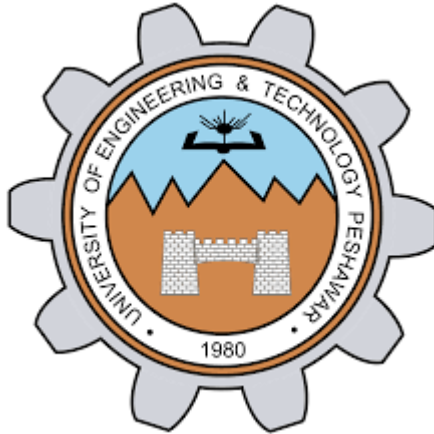




Lab Report – EE-170L Computer Programming (Lab 02)

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1. Objectives

- Understand the concept and syntax of the **for loop** in C++.
- Learn how to implement **nested for loops** for structured and repeated operations.
- Apply loops to solve problems such as counting, pattern printing, and mathematical computations.

2. Theory

2.1 The for Loop

A for loop is a control structure that allows code to be executed repeatedly based on a condition. It is typically used when the number of iterations is known beforehand.

Syntax:

```
for (initialization; condition; update) {  
    // statements to execute  
}
```

initialization: Executed once at the beginning. Used to initialize loop control variables.

- Condition: Evaluated before each iteration. If true, the loop body executes.
- Update: Executed after each iteration. Typically used to modify the loop variable.

Flow Diagram:

Code

Start

|

Initialization

|

Condition True? ---- No ---> Exit Loop

|

v

Statements

|

Update

|

Go back to Condition

2.2 Nested for Loops

A nested loop is a loop inside another loop. The inner loop completes all its iterations for each iteration of the outer loop. Useful for multi-dimensional data like tables and patterns.

3. Programs Implemented

3.1 Countdown from 5 to 1

```
#include <iostream>          // Include input-output stream
using namespace std;        // Use standard namespace

int main() {                // beginning of main function
    for (int i = 5; i > 0; i--) { // Initialize i=5; loop while i>0; decrement i
        cout << i <<endl; // Print current value of i
    }
    cout << "outside loop now"; // Print after loop ends
    return 0;                // End of program
}
```

Output:

```
5
4
3
2
1
outside loop now
```

3.2 Prime Number Check

```
#include <iostream>
using namespace std;
```

```

int main() {

    int num;                // Variable to store user input

    cout << "Enter a number please\n";    // Prompt user

    cin >> num;              // Read input

    int flag = 0;           // Flag to track if number is not prime

    for (int i = 2; i < num; i++) {    // Loop from 2 to num-1
        if (num % i == 0) {            // Check if num is divisible by i
            flag = 1;                  // Set flag if divisible
            break;                     // Exit loop early
        }
    }

    if (flag == 1) {
        cout << num << " is not a prime number\n";    // Not prime
    } else {
        cout << num << " is a prime number\n";    // Prime
    }

    return 0;
}

```

3.3 Nested Loop Pattern Printing

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    for (int num = 7; num > 0; num--) {    // Outer loop from 7 to 1
```

```
        for (int j = 1; j <= num; j++) {    // Inner loop from 1 to num
```

```
            cout << j << " ";    // Print numbers in row
```

```
        }
```

```
        cout << "\n";    // New line after each row
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

```
1 2 3 4 5 6 7
```

```
1 2 3 4 5 6
```

```
1 2 3 4 5
```

```
1 2 3 4
```

```
1 2 3
```

```
1 2
```

```
1
```

4. Lab Tasks

4.1 While Loop: Display Numbers from 1 to 10

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int i = 1;           // Initialize counter
```

```
    while (i <= 10) {    // Loop while i <= 10
```

```
        cout << i << endl;    // Print current number
```

```
        i++;               // Increment counter
```

```
    }
```

```
    return 0;
```

```
}
```

4.2 Multiplication Table using Nested For Loops

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```

for (int i = 1; i <= 10; i++) {    // Outer loop for rows
    for (int j = 1; j <= 10; j++) { // Inner loop for columns
        cout << i * j << "\t";    // Print product with tab
    }
    cout << endl;                // New line after each row
}
return 0;
}

```

4.3 Sum of Numbers from 1 to N

```

#include <iostream>    // include input/output library
using namespace std; // use standard namespace

int main() {
    int n, sum = 0;    // declare n and initialize sum
    cout << "Enter a positive integer: "; // ask user for input
    cin >> n;          // read input into n

    for (int i = 1; i <= n; i++) { // loop from 1 to n
        sum += i;                // add i to sum
    }
}

```



```
cout << "Sum = " << sum << endl; // display result  
return 0;    }           // end program
```

4.4 Triangle Pattern Printing

```
#include <iostream>  
  
using namespace std;  
  
int main() {  
    int n, sum = 0;           // Declare variables  
    cout << "Enter a positive integer: "; // Prompt user  
    cin >> n;                 // Read input  
  
    for (int i = 1; i <= n; i++) { // Loop from 1 to n  
        sum += i;               // Add i to sum  
    }  
  
    cout << "Sum = " << sum;    // Display result  
    return 0;  
}
```

5. Conclusion

- The lab provided hands-on experience with **for loops**, **while loops**, and **nested loops**.
- Students learned how to control program flow using loop constructs.
- Practical applications included counting, checking prime numbers, generating multiplication tables, and printing patterns.
- These concepts are foundational for solving repetitive tasks efficiently in programming.